

# Strong Artificial Intelligence

F Action

Changer

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# Repetition „Information“

- What is information?
- Where is information?
- How is information transferred from sender to receiver?
- Can information be deleted and how?

# F Action

- |                  |                  |
|------------------|------------------|
| 1. Actor         | Mover            |
| 2. Physical Act  | Move Things      |
| 3. Mental Act    | Move Thoughts    |
| 4. Speech Act    | Move Information |
| 5. Action Theory | Component Two    |
| 6. Action Plan   | Schema           |
| 7. Conclusion    | Part of Core     |

Intelligence  
Ability [of human beings], to **think**  
**abstract** as well as **reasonable** and  
derive **purposeful acting** out of that

# Workshop „Action“

1. What is an action? Define that as precise and as abstract as possible. Think also of the work matrix we used in chapter A.
2. Who can execute actions? Or what is needed to execute actions?
3. Which different types and purposes of actions would you distinguish?
4. Do you think, an action is the only connection between the mental and the physical world? Can you find other connections?

# 1 Actor

Mover



## 1. Actor

### Prerequisites

1. must be able to move



2. must be able to think

- no purposeful acting without anticipation
- planing means abstracting
- learn



3. must be able to communicate  
(except lone wolves)



## 1. Actor

### Three Action Types

outwards

inwards



# 1. Actor

## Expression



by body



by speech

Physical Entities

Mental Entities

Information





# 2 Physical Act

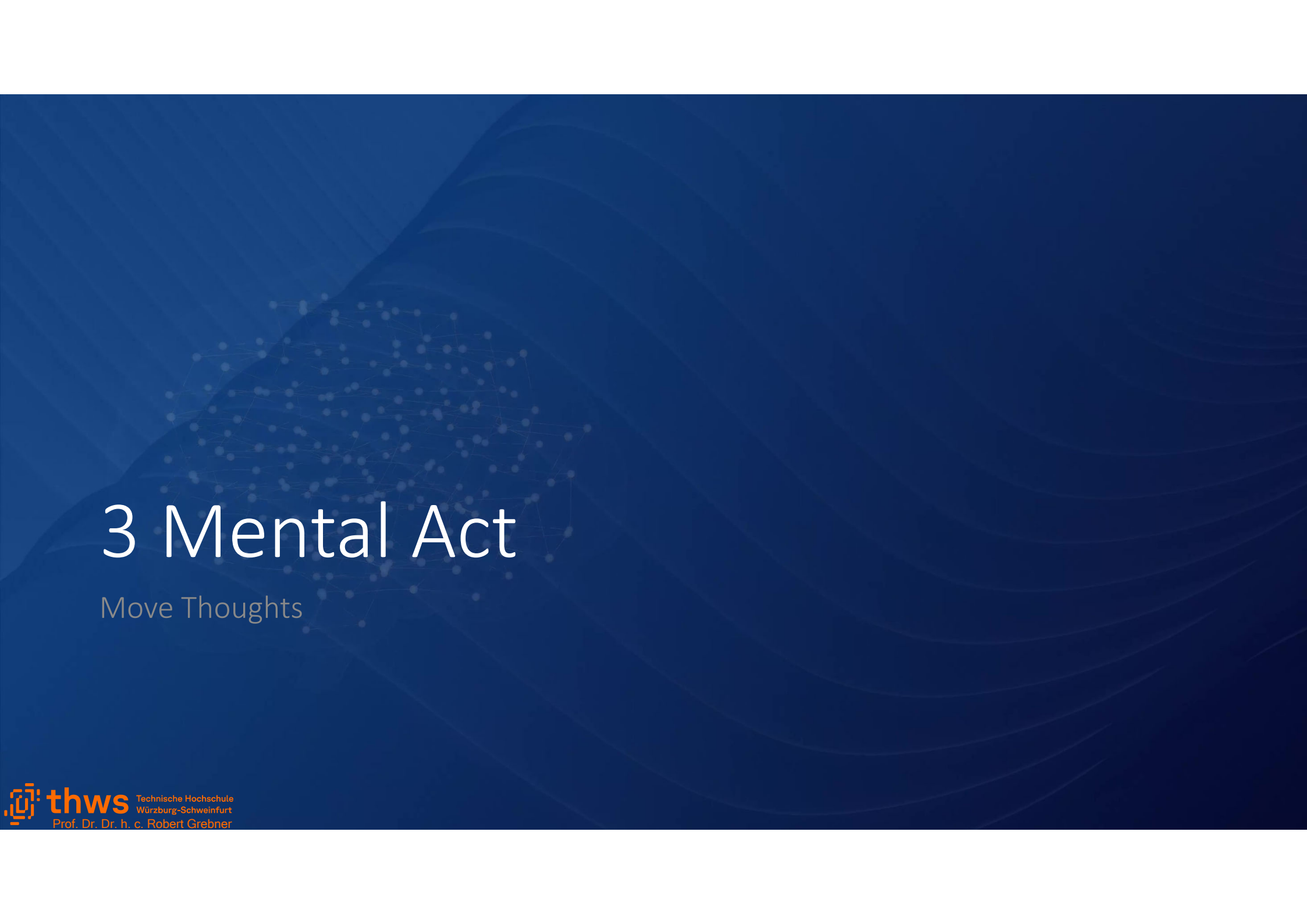
Move Things

## 2. Physical Act



### Purpose

- 1.a.) Physical: Change own Position
- 1.b.) Mental: Change others Mind  
(expression – body language)  
(set or generate information)
- 1.c.) Physical and Mental: Change  
Position to set/get Information
- 2.a.) Physical: Change others Position
- 2.b.) Mental: Change others Mind  
(expression)  
(set or generate information)
- 2.c.) Physical and Mental: Change  
Position to set/get Information



# 3 Mental Act

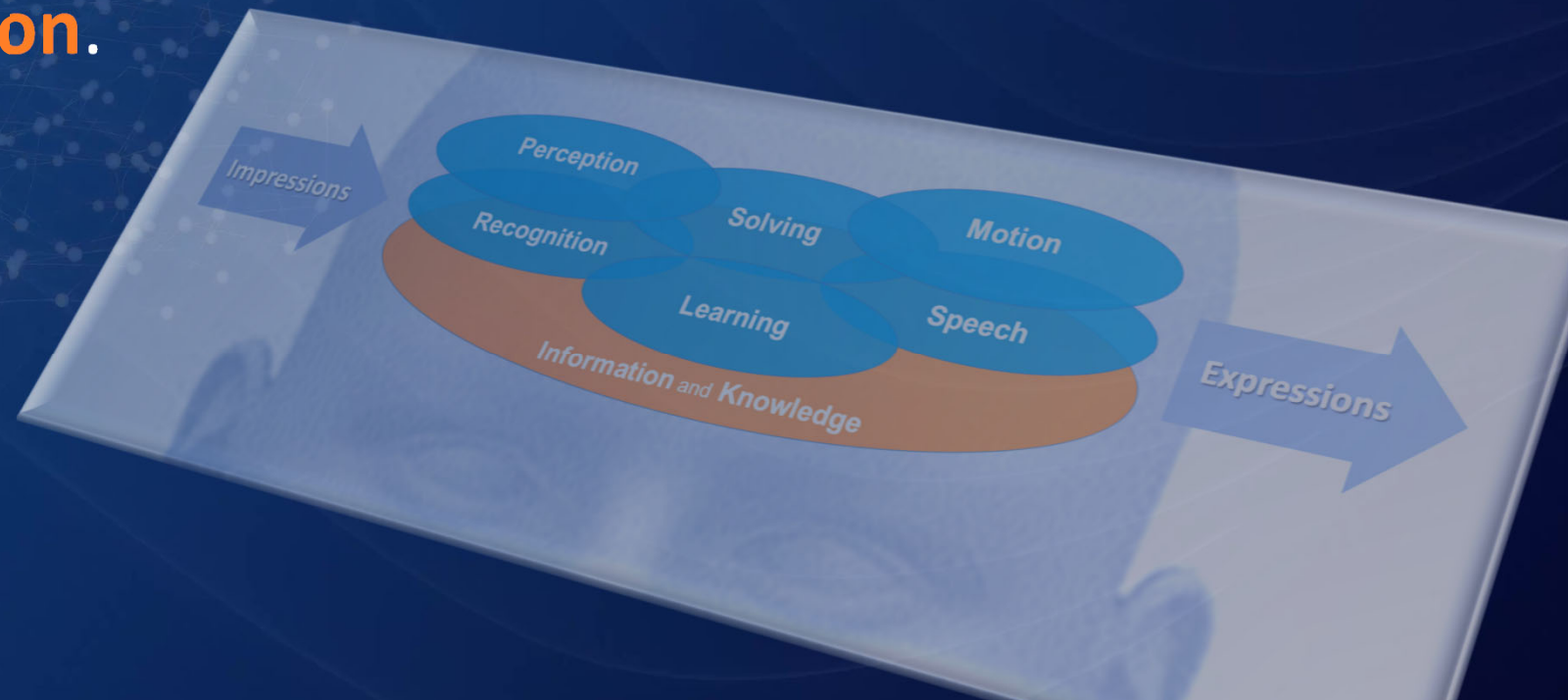
Move Thoughts

# Workshop „Thinking Act “

1. Why is thinking an action? What is thinking doing?
2. Which „tools“ are used while thinking?
3. What is changed by thinking?
4. How can we determine a need for action?

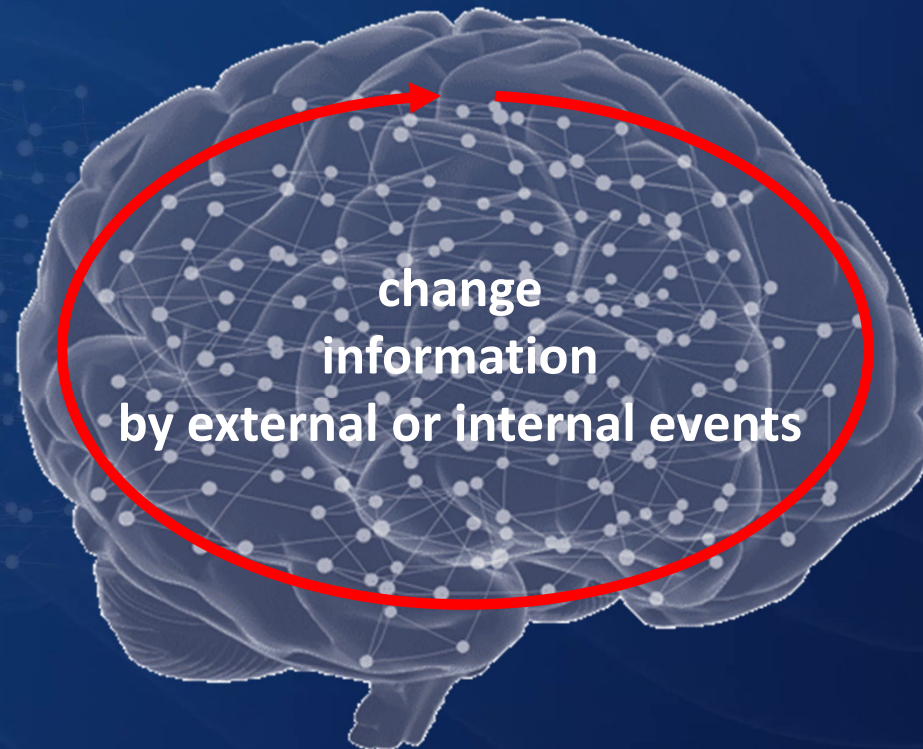
## 4. Thinking Act

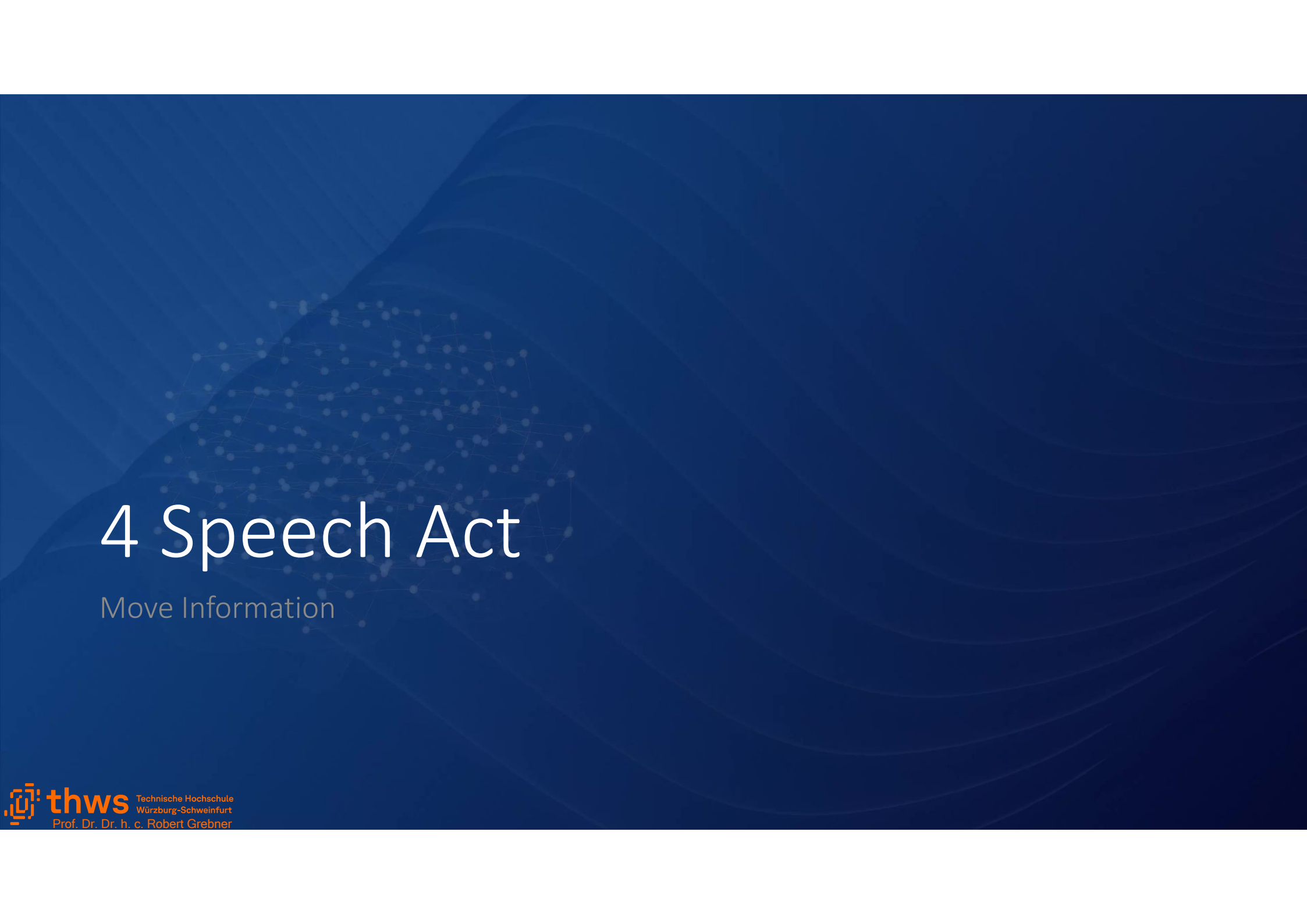
Processing of impressions, solving challenges, learning, planing and executing **purposeful ACTIONS** after determing a **need for action**.





## 4. Thinking Act





# 4 Speech Act

Move Information

# Workshop „Speech Act “

1. How can speaking be an action?
2. What is changed during a speech?
3. Which types of sentences do you know?
4. Can speaking move physical entities? And if yes, how?

### 3 Speech Act

1.)   thoughtful

2.)   rash

#### Purpose

- 1.a.) Change information of another Individuals by a declaration.
- 1.b.) Get information from another Individual by a question.
- 1.c.) Physical: Order an individual to manipulate physical entities
- 2.a.) Quick reaction to upsetting.
- 2.b.) Quick reaction to physical stimuli.
- 2.c.) Quick reaction to optical stimuli.



# 5 Action Theory

## Component Two



# Workshop „Action Theory“

1. Which questions should an action theory answer?
2. How can action theory be connected to an information theory?
3. How can it be connected to a thought theory?

## 5. Action Theory

### Thesis

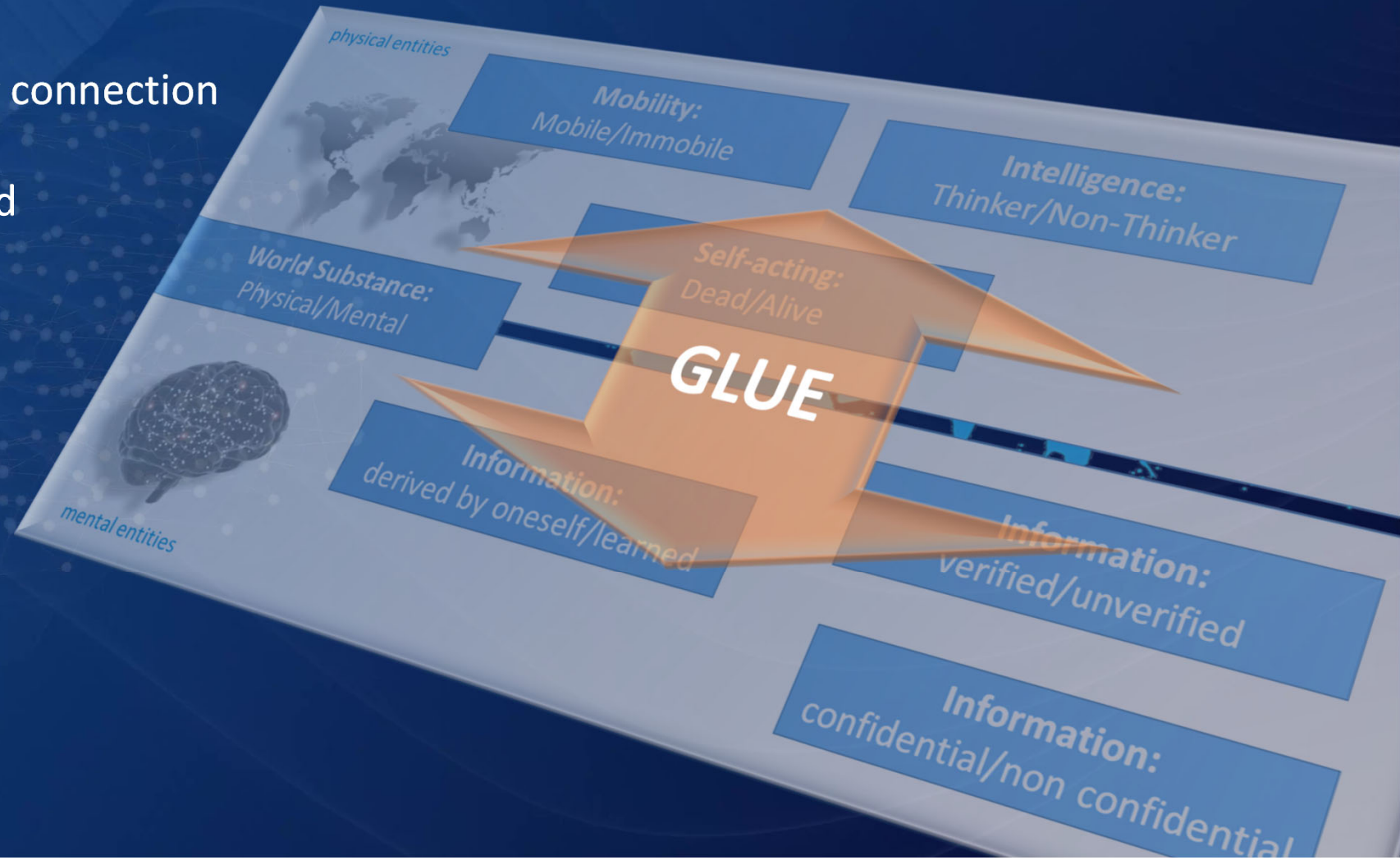
- What is this theory needed for?
  - to explain, why individuals do what they do
  - in the social domain it is for explaining how persons can live and work together
- What is an action?
  - Change in the physical or mental world
  - Actions always create, change or delete entities
  - Action always is combined with motion/movement
- Why and when is an action started?
  - When perceived impressions do not correspond to my inner values
- Actions are framed by a system we already know: a kind of work system



## 5. Action Theory

### Purpose of action

Action is the only connection  
between  
the physical world  
and  
the mental world



## 5. Action Theory

### Fundamentals

What is Action?

Manipulation of Entities.

How is the Need for Action  
Determined?

When Perceived Impressions  
do not Meet Values.

Who Decides about the  
Necessity of Manipulation?

The Brain.

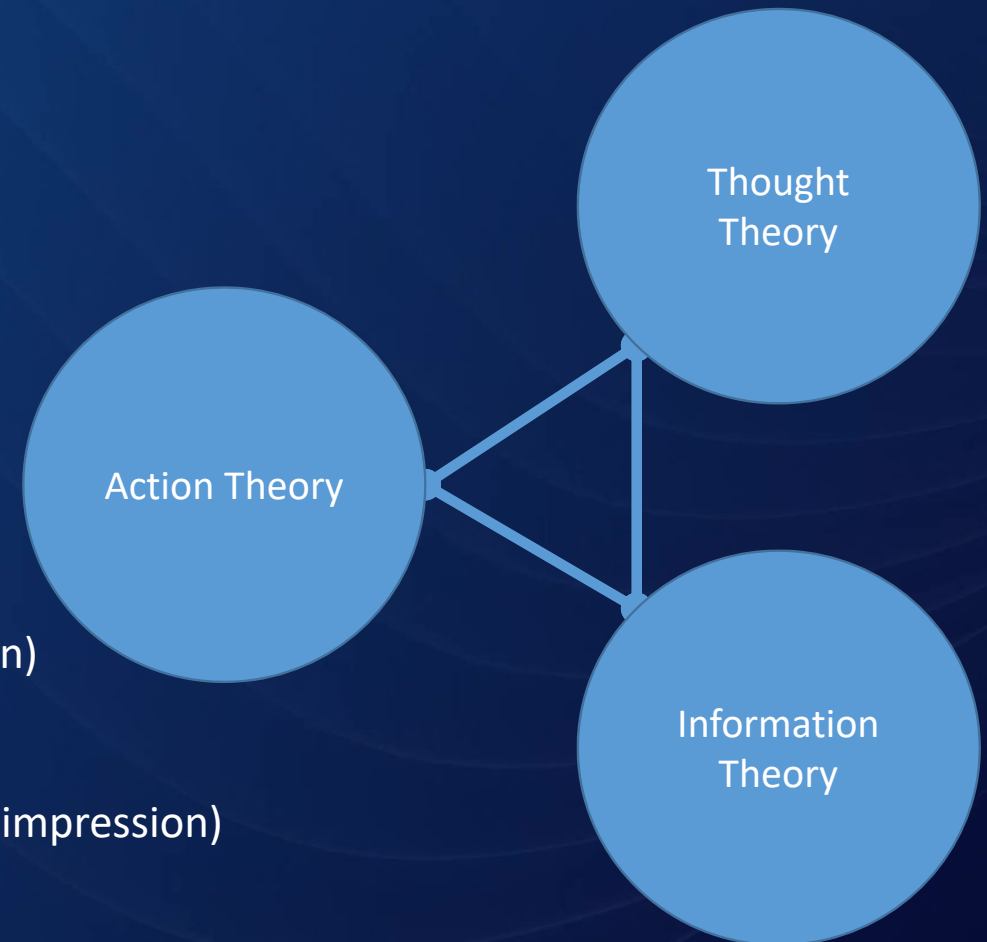
What is to be Done before  
Acting?

Producing an Action Plan.

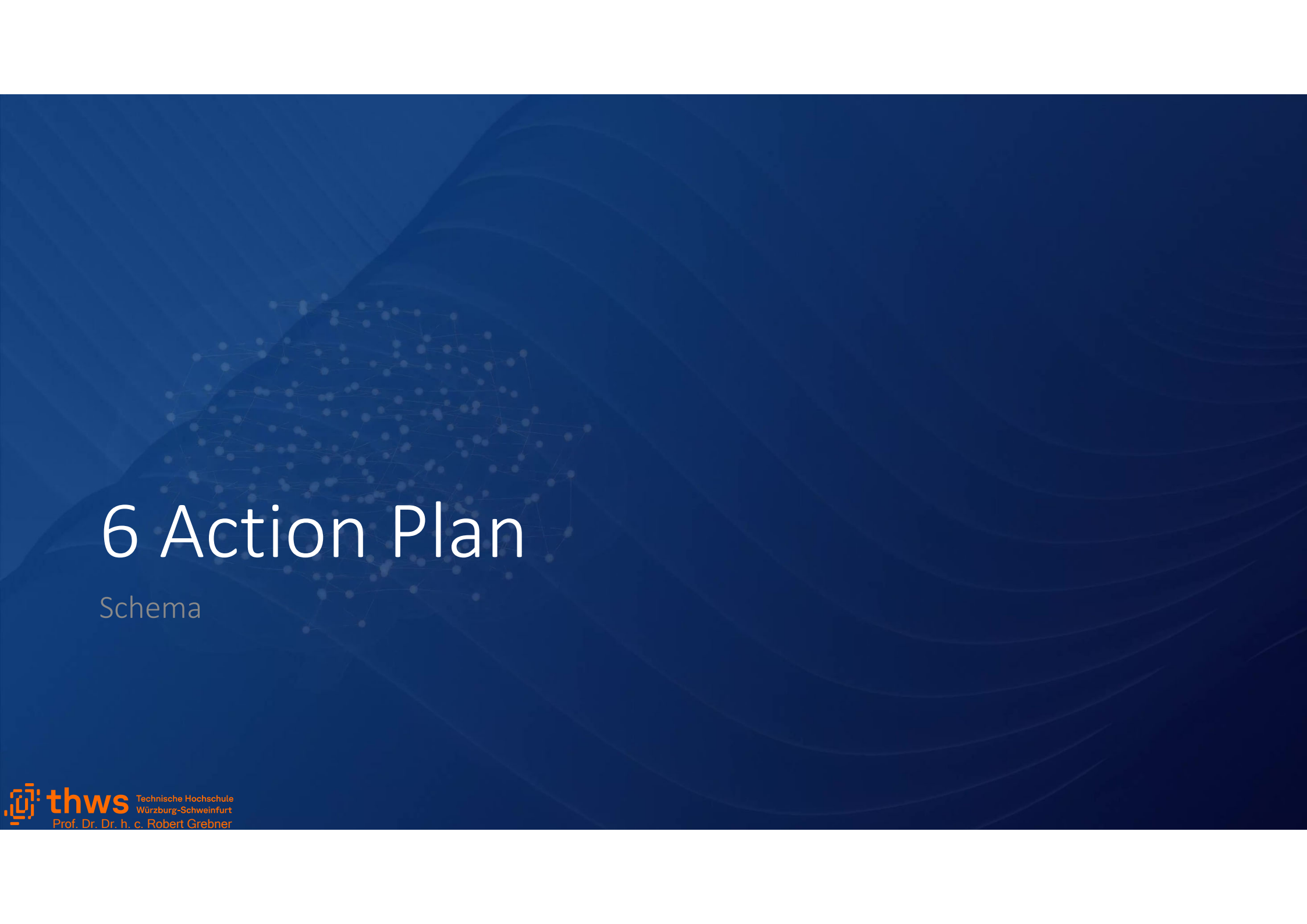
## 5. Action Theory

Part of three phenomena

- Thinking generates information
- Information is what thinking is working on
- Action is what thinking drives
- Action is the output of thinking
- Action is what can be seen from thinking (expression)
- Thinking is massively influenced by action
- Information is what will be produced out of action (impression)
- Action is the source of information







# 6 Action Plan

Schema



# Workshop „Action Plan“

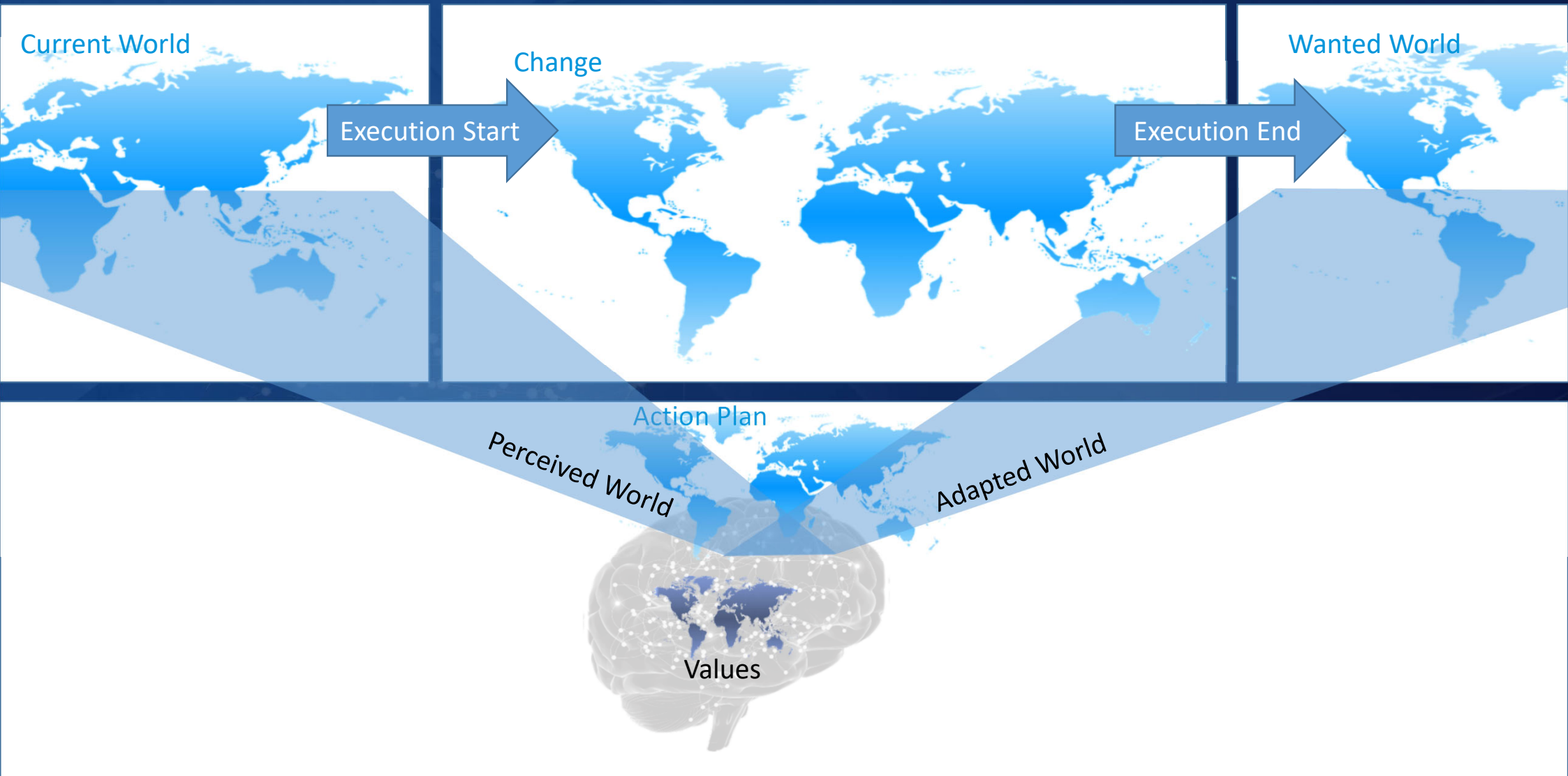
Think of a specific situation, where you changed a small thing in the world (for example leaving the table tennis game to come back to the lecture).

1. What values were not fulfilled?
2. From what was the action plan derived?
3. What was the content of the action plan?

- Actions are always based on a plan.
- A plan is produced in our mind by thinking.
- A plan shows how to transform a discourse world from an initial state to a final state.
- All single steps of motion and of speech are part of the action plan.
- The processing of the plan is carried out by motion and speech.

## 6. Action Plan

## Planing and Execution





# 7 Conclusion

Part of Core

## 7. Conclusion

- Proper action needs planing and anticipating (thinking into the future) .
- Action covers three action areas: Physical act, speech act and thinking act.
- Action theory gives an explanation of why action is started.
- Action plans are the base of implementing or executing actions.
- Action is the core of intelligence. It combines all aspects of the brain together.
- Action is closely connected to information and thinking.





I  
Thank  
You